

Post-Disaster Rescue Guidance via VLM Fine-Tuning

CSE499A – Section 15, Group 5, Update 2

Name	Student ID
Abdullah Al Noman	2022095042
Tamanna Akter Mou	2211951042
Aryan Sami	2231407042
Ridita Afrin Riya	2211622042
Abrar Mohammed Tanzim Alam	2222864042

Weekly Update:

Our primary objective this week was to conduct a rigorous and comprehensive literature review phase to position our proposed framework, *Post-Disaster Rescue Guidance via Fine-Tuning Vision Language Models*.

Work Done by Abrar Mohammed Tanzim Alam

- Literature Review & Presentation on DisasterM3 (Wang et al., NeurIPS 2025):** Analyzed and presented the DisasterM3 benchmark—the first global-scale remote sensing VLM dataset unifying multi-hazard (10 types), multi-sensor (optical + SAR), and multi-task evaluation: 26,988 bi-temporal images, 123,010 instruction pairs, 36 events, 5 continents. Mapped its 9-task taxonomy (recognition → quantification → localization → reporting → relational reasoning) and hybrid Human-AI ground truth pipeline (GPT-4o prompts → expert annotation → GPT-4o polishing → multi-round verification). Fine-tuning Qwen2.5-VL-7B/InternVL3-8B on 92K pairs yielded +10.4% QA accuracy, +2.1 report score, +40.8% segmentation mIoU. Identified limitations (counting overfitting, optical-vs-SAR gaps) and mapped relevance: their GPT-4.1 rubric validates our RAR metric, and their Qwen2.5-VL + LoRA architecture is structurally identical to our QLoRA pipeline.
- Unified AIDER Processing Pipeline ([support/aider_pipeline.ipynb](#)):** Built an end-to-end notebook unifying three stages that closely mirror DisasterM3’s dataset construction pipeline, directly addressing its 9-task taxonomy (scene/disaster-type/bearing-body recognition, restoration advice), on a small image pool for validation before full-scale execution; beneficial techniques will be merged into our production pipeline. Initializes `WORKER_ID` sharding (5 members), multi-key Gemini API rotation (`gemini-3.5-flash` → `gemini-2.5-flash`), and class exclusion filters. Stage 1 standardizes 6,433 images (RGB conversion, 1280px cap, sub-256px rejection, MD5 dedup, resume-safe tracker).
- Synthetic Metadata Generation (Stage 2):** Queries Gemini to synthesize pseudo-xBD JSON metadata per image—1–5 instance-level features with damage subtypes (`destroyed/major-damage/minor-damage/no-damage`). Uses class-balanced quotas, worker-level sharding, JSON validation, and a retry engine handling 503/429 errors with key rotation and model fallback.
- Tactical Ground Truth Augmentation (Stage 3):** Combines metadata + imagery into rescue instruction-response pairs across four perspectives (`structural`, `triage`, `logistics`, `environmental`) with disaster-specific prompts. Enforces QA gates (3-section schema, 120+ words, first-person rejection) and perspective-based deduplication, producing 117 verified pairs to `dataset/aider_train_dataset.worker_0.json`.

Work Done by Tamanna Akter Mou

- Literature Review on Longitudinal Remote Sensing:** Performed an in-depth academic review of a high-impact remote-sensing paper mapping satellite monitoring actions over 95 unique global cultural heritage sites affected by earthquakes, floods, and storms.
- Technical Performance Benchmarking:** Analyzed empirical tracking configurations where traditional CNN systems increased regional post-disaster classification metrics from 75% to 95% across a five-year study framework.
- Research Gap Extraction:** Documented a severe systemic limitation in current remote-sensing setups: most frameworks remain passive and historical, lacking conversational interactivity, text-based querying, or instant tactical command generation for rescue teams.
- Relevance Mapping to Our Pipeline:** Connected these limitations to our proposed VLM workflow by showing why a disaster model should not only classify scenes but also convert visual evidence into actionable natural-language guidance.

Work Done by Abdullah Al Noman

1. **Literature Review on Multimodal Vision-Language Foundations:** Conducted a thorough analysis of modern vision foundation pipelines, including InternVL-style and LLaVA-style systems, investigating how spatial visual features align with token spaces through projection layers.
2. **Generative Assessment Metric Analysis:** Critiqued standard automated text-generation evaluations such as ROUGE-L and BERTScore, documenting their failure to account for rescue accuracy, navigational hazards, or immediate structural safety constraints.
3. **Metric System Alignment:** Documented how the group’s custom Rescue Actionability Rubric (RAR) addresses these baseline gaps by penalizing safety-critical hallucinations, including missing exit paths, misidentified debris blockages, or unsafe route recommendations.
4. **Framework Justification:** Summarized how multimodal alignment research supports our transition from static disaster recognition toward instruction-following VLM outputs for field-level decision support.

Work Done by Aryan Sami

1. **Literature Review on Medium-Resolution Satellite Damage Assessment:** Analyzed the Copernicus-based building damage assessment framework proposed by Dietrich et al. (2026), which evaluates U-Net, ChangeMamba, and pretrained geospatial foundation models (Prithvi-EO-2.0, DOFA) on Sentinel-1 SAR and Sentinel-2 optical imagery, benchmarked against the xBD/xView2 VHR damage-labeling standard.
2. **Resolution and Localization Gaps Identified:** Documented how 10 m ground sampling distance limits fine-grained damage detection, particularly for sparse, low-intensity damage in dense urban settings (e.g., the Mexico earthquake failure case), and how building localization, not damage classification, remains the weakest stage of the pipeline at this resolution, constraining how reliably free satellite data can stand in for VHR imagery.
3. **Multi-Source Data Integration Rationale:** Outlined how combining pre-/post-event optical (Sentinel-2) and radar (Sentinel-1) imagery, rather than relying on a single source, improves robustness to cloud cover and lighting differences, and how this multi-source approach directly supports generating more reliable, damage-aware rescue guidance from disaster imagery.
4. **Dataset-to-Model Mapping:** Mapped the xBD-S12 dataset structure, spatiotemporally aligned Sentinel-1/Sentinel-2 patches paired with corrected xBD building-damage labels, against our rescue-oriented instruction-response format, clarifying how pre/post satellite comparisons and damage classifications can be converted into training-ready multimodal examples for generating actionable rescue instructions.

Work Done by Ridita Afrin Riya

1. **Literature Review on Efficiency Adapters:** Conducted a comprehensive review of academic papers focused on parameter-efficient low-rank adaptation methods, tracking performance changes across 4-bit and 8-bit base-weight freezing strategies.
2. **Hardware Constraint Verification:** Documented memory optimization results showing how QLoRA-style layers can approximate full-parameter optimization while reducing target GPU memory utilization footprints by up to 60%.
3. **Infrastructure Mapping:** Aligned efficiency benchmarks with our local cluster and Kaggle configurations to confirm that the Qwen2-VL-7B fine-tuning workflow with Unsloth can operate smoothly within restricted T4 VRAM allocations.
4. **Fine-Tuning Readiness Notes:** Prepared implementation notes linking adapter rank, quantization settings, batch-size limits, and expected checkpoint storage requirements for the upcoming training phase.

***Next Week’s Plan:** We will broaden the literature review by analyzing additional recent research papers on multimodal disaster architectures and fine-tuning optimizations. Concurrently, we will integrate insights from all reviewed works into our existing pipeline, so as to come up with a targeted solutions within our framework before proceeding to the model fine-tuning.